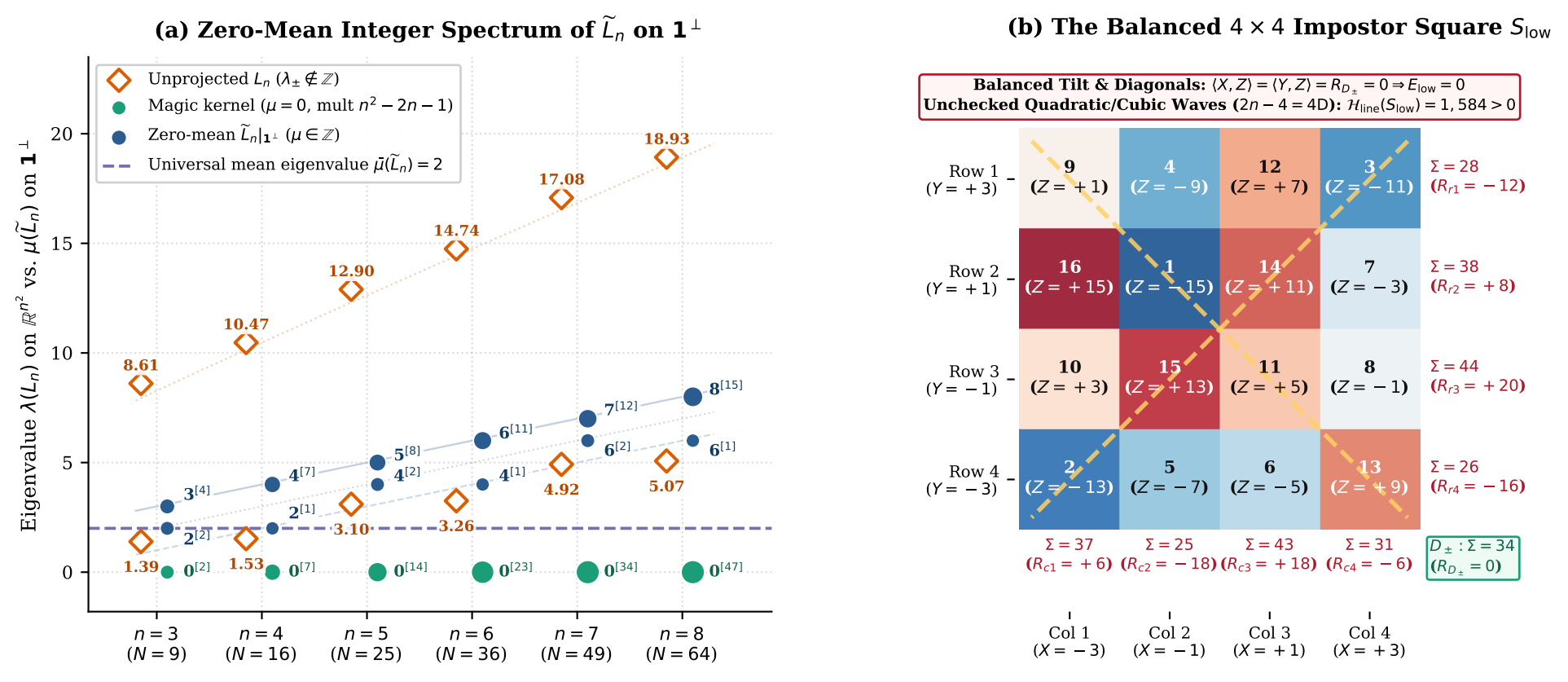

(a) Zero-Mean Integer Spectrum of $\widetilde{L}_n$ on $\mathbf{1}^\perp$
Unprojected L_n ($\lambda_\pm \notin \mathbb{Z}$)
Magic kernel ($\mu = 0$, mult $n^2 - 2n - 1$)
Zero-mean $\widetilde{L}_n|_{\mathbf{1}^\perp}$ ($\mu \in \mathbb{Z}$)
Universal mean eigenvalue $\bar{\mu}(\widetilde{L}_n) = 2$
Eigenvalue $\lambda(L_n)$ on $\mathbb{R}^{n^2}$ vs. $\mu(\widetilde{L}_n)$ on $\mathbf{1}^\perp$
8.61
10.47
12.90
14.74
17.08
18.93
1.39
1.53
3.10
3.26
4.92
5.07
$n = 3$ ($N = 9$)
$n = 4$ ($N = 16$)
$n = 5$ ($N = 25$)
$n = 6$ ($N = 36$)
$n = 7$ ($N = 49$)
$n = 8$ ($N = 64$)
(b) The Balanced 4×4 Impostor Square S_{low}
Balanced Tilt & Diagonals: $\langle X, Z \rangle = \langle Y, Z \rangle = R_{D_\pm} = 0 \Rightarrow E_{\text{low}} = 0$
Unchecked Quadratic/Cubic Waves ($2n - 4 = 4\text{D}$): $\mathcal{H}_{\text{line}}(S_{\text{low}}) = 1,584 > 0$
Row 1 ($Y = +3$)
Row 2 ($Y = +1$)
Row 3 ($Y = -1$)
Row 4 ($Y = -3$)
9 ($Z = +1$)
4 ($Z = -9$)
12 ($Z = +7$)
3 ($Z = -11$)
16 ($Z = +15$)
1 ($Z = -15$)
14 ($Z = +11$)
7 ($Z = -3$)
10 ($Z = +3$)
15 ($Z = +13$)
11 ($Z = +5$)
8 ($Z = -1$)
2 ($Z = -13$)
5 ($Z = -7$)
6 ($Z = -5$)
13 ($Z = +9$)
$\Sigma = 28$ ($R_{r1} = -12$)
$\Sigma = 38$ ($R_{r2} = +8$)
$\Sigma = 44$ ($R_{r3} = +20$)
$\Sigma = 26$ ($R_{r4} = -16$)
$\Sigma = 37$ ($R_{c1} = +6$)
$\Sigma = 25$ ($R_{c2} = -18$)
$\Sigma = 43$ ($R_{c3} = +18$)
$\Sigma = 31$ ($R_{c4} = -6$)
$D_\pm : \Sigma = 34$ ($R_{D_\pm} = 0$)
Col 1 ($X = -3$)
Col 2 ($X = -1$)
Col 3 ($X = +1$)
Col 4 ($X = +3$)